

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

Answers will vary, but all traffic from R3 to the 192.168.2.0/24 network will be routed through R1. The S0/0/1 interface on R2 remains configured as a passive interface, so OSPF routing information is not advertised across this interface. The accumulated cost of 129 results from the fact that traffic from R3 to the 192.168.2.0/24 network must pass through two T1 (1.544 Mb/s) serial links (each with a cost of 64) plus the R2 Gigabit 0/0 LAN link (with a cost of 1).

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1.

R2(config-router)# no passive-interface s0/0/1.

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 (One T1 (1.544 Mb/s) serial link (at a cost of 64) plus the R2 Gigabit 0/0 LAN link (at a cost of 1)).

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

Please ensure reference bandwidth is consistent across all routers.

Why would you want to change the OSPF default reference-bandwidth?

Answers may vary, but today's networking equipment supports link speeds exceeding 100 Mb/s. To achieve more accurate cost calculations for these faster links, a higher default reference bandwidth is required.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24: R1 S0/0/1 + R3 G0/0 (781+1=782). The cost to 192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1 (781+64=845).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1,562. Each serial link now has a cost of 781, and the route to the 192.168.23.0/24 network travels over two serial links. $781 + 781 = 1,562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF will choose the route with the least accumulated cost. The route with the lowest accumulated cost is: R1-S0/0/0 + R2-S0/0/1 + R3-G0/0, or $781 + 781 + 1 = 1,563$. This metric is smaller than the accumulated cost of R1-S0/0/1 + R3-G0/0, or $1565 + 1 = 1,566$.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

In multi-access networks, the election of the Designated Router (DR) and Backup Designated Router (BDR) is governed by each device's Router ID. When this ID comes from a regular physical interface, it may shift if that interface ever fails. To always keep the Router ID stable, administrators should either assign it from the IP address of a loopback interface—which stays up permanently—or configure it directly with the router-id command.

2. Why is the DR/BDR election process not a concern in this lab?

The selection of a Designated Router (DR) and its Backup Designated Router (BDR) apply exclusively to multi-access network topologies, such as Ethernet and Frame Relay. In this lab setup, however, the serial links are configured strictly as point-to-point connections; as a result, the DR/BDR selection mechanism is skipped entirely.

3. Why would you want to set an OSPF interface to passive?

Answers may vary, but placing a LAN interface in passive mode prevents the router from sending unnecessary OSPF routing messages across that link, thereby conserving bandwidth. Nevertheless, the router will continue to advertise the connected network to its OSPF neighbors.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF achieves its hierarchical structure and exceptional scalability by dividing the entire routing domain into multiple independent areas. This area-based organization confines link-state updates within each area, dramatically reducing routing overhead and enabling the protocol to support extremely large networks without performance degradation.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

The single command required is `network 10.0.0.0 0.255.255.255 area 0` (Entered in router configuration mode, this statement activates OSPF Area 0 on every interface whose IP address falls inside the 10.0.0.0/8 range.)

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The route provided by EIGRP will be installed in the routing table and used for the forwarding decision. When the same destination prefix is learned from multiple routing protocols, the router first compares Administrative Distance (AD) values rather than metrics. EIGRP's default AD of 90 is lower than OSPF's 110 and RIP's 120; therefore, the EIGRP route is preferred regardless of the metric values shown.